

Fatemeh Zarinjoui

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Education

Master in Computer Science

2023 – Present

Sharif University of Technology, Tehran

Ranked 1/66 in the Department of Mathematical Sciences; GPA: 19.67 / 20

Thesis: Zero-Knowledge Machine Learning (ZKML)

Bachelor in Computer Science

2019 – 2023

University of Isfahan, Isfahan

Ranked 1/56 in the Computer Science Department; GPA: 17.87 / 20

Research Interests

My research interests are in applied cryptography, with a particular focus on using the provable security methodology to build practical, privacy-preserving systems for the real world. I am also deeply interested in emerging challenges, such as the development of post-quantum cryptography to ensure long-term security for real-world applications and internet protocols.

Research Experience

ZKRNN: Zero-Knowledge Proofs for Recurrent Neural Networks

2024 – Present

Graduate Research Assistant under the supervision of **Dr. Behzad Abdolmaleki & Dr. Shahram Khazaei**

The University of Sheffield, South Yorkshire, England & Sharif University of Technology, Tehran, Iran

- Designing cryptographic protocols to generate succinct zero-knowledge proofs for Recurrent Neural Networks (RNNs) using Incrementally Verifiable Computation (IVC).
- The resulting system achieves linear prover time with a succinct verifier and proof size that remains independent of the number of time steps. (**To be submitted to ACM CCS 2026**, Manuscript is available upon request).

Secure E-Voting with Everlasting Privacy

2025

Research Intern under the supervision of **Prof. Bryan Ford**

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

- Trust-Limited Coercion-Resistant (protects against forced voting) In-Person Voter Registration toward post-quantum everlasting privacy in e-voting.
- Leverage information-theoretic assumptions to resist future adversaries in democratic systems.

Proof-of-Personhood Usability Study for Digital Democracy

Summer 2025

Research Intern under the supervision of **Prof. Bryan Ford**

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

- Developed a mobile-friendly web application for Proof-of-Personhood (PoP) to support usability evaluation in digital democracy contexts.
- Conducted a pilot study to test the system and gather preliminary feedback; currently in the process of obtaining HR approval for the full-scale user study.

Privacy-Preserving Federated Learning

2023 – 2024

Graduate Research Assistant under the supervision of **Dr. Shahram Khazaei**

Sharif University of Technology, Tehran, Iran

- Studied privacy-preserving mechanisms for FL, focusing on secure aggregation and MPC techniques.
- While developing experience in applied cryptography, I identified a flaw in an IEEE TIFS 2023 verifiable aggregation protocol and proposed a fix to restore completeness ([Manuscript](#)).

Automated Quality Control using YOLO for Electronic Boards

2022 – 2023

Undergraduate Research Project under the supervision of Dr. Mohsen Alambardar

University of Isfahan, Isfahan, Iran

- Implemented a YOLO-based pipeline to inspect boards on a conveyor belt, detecting missing/misplaced components.
- Deployed on Raspberry Pi for industrial QA and manufacturing integration.

Selected Teaching Experience

Teaching Assistant — Introduction to Cryptography (Graduate)

2024

Dr. Shahram Khazaei, Department of Mathematical Sciences, Sharif University of Technology

Teaching Assistant — Foundation of Blockchains (Graduate)

2023

Prof. Mohammad Ali Maddah-Ali, Dept. of Electrical & Computer Eng., University of Minnesota, Twin Cities

Teaching Assistant — Mathematical Database

2023

DR. Mojtaba Rafiee, Mathematics and Computer Science Department, University of Isfahan

Head Teaching Assistant — Data Structures and Algorithms

2022

DR. Jafar AlmasiZadeh, Mathematics and Computer Science Department, University of Isfahan.

Relevant Courses

Topics in Cryptography	20/20	Design and Analysis of Algorithms	20/20
Theory of Distributed Systems	20/20	Principles of Operating Systems	20/20
Introduction to Cryptography	20/20	Principles of Network Security	19.75/20
Machine Learning Theory	20/20	Matrix Computation	20/20
Advanced Programming	20/20	Linear Optimization	20/20

Selected Course Projects

A Survey on Byzantine Fault Tolerance Algorithms

Dec 2024

Theory of Distributed Systems course project, Dept. of Mathematical Sciences, Sharif University of Technology

Computer Vision Project

Dec 2023

Machine Learning Theory course project, Department of Mathematical Sciences, Sharif University of Technology

Face Recognition using Eigenfaces & PCA

May 2023

Matrix Computations course project, Department of Mathematical Sciences, Sharif University of Technology

Implementation of RSA From Scratch

Jan 2020

Advanced Programming Project, Mathematics and Computer Science Department, University of Isfahan

Selected Talks

Federated Learning Meets Multi-objective Optimization

Jun 2024

Optimization Seminars, Sharif University of Technology

Nearest Neighbor Search Methods

Jan 2024

Computer Science Seminars, Sharif University of Technology

Outlier Detection

Jan 2024

Machine Learning Seminars, Sharif University of Technology

Skills

Programming Languages: Python, C++, GO, SQL, JavaScript, Solidity, RISC-V, HTML/CSS, Bash

Machine Learning: PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, YOLO

Tools: $\LaTeX$, Git, Linux, PostgreSQL

Volunteer Service

Active Member, Academic Association of Mathematics and Computer Science

2022

- Organized academic and social events.
- Contributed articles to the association's student journal.
- mentoring undergrads to find their field of expertise

Languages

Persian: Native

English: Fluent | *TOEFL: 107*